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Case Report

Mandibular Osteomyelitis as a Complication of Ludwig's Angina – Case Report

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ABSTRACT

Introduction: Ludwig's angina is a rapidly progressing gangrenous cellulitis that involves the submandibular, sublingual, and submental spaces. Its standard management is based on three pillars: securing the airway, broad-spectrum antibiotic therapy, and surgical drainage. The coexistence of simultaneous active infectious foci—a deep submandibular abscess and mandibular osteomyelitis—along with resolution of the abscess without surgical drainage constitutes an exceptional presentation that is rarely documented in the medical literature.

Clinical Case: A 48-year-old male patient with a history of hypertension presented with a one-week history of left lower mandibular pain, a left submandibular mass, odynophagia, and purulent discharge at the base of the tongue, with progressive deterioration leading to acute respiratory failure that required orotracheal intubation. Ludwig's angina with an abscess causing airway collapse was diagnosed, along with mandibular osteomyelitis confirmed identified intraoperatively during dental extraction and debridement. Intravenous antibiotic therapy was indicated, along with conservative management without surgical drainage of the submandibular abscess. Progressive resolution of the submandibular abscess was observed under treatment, with successful extubation on the sixth day of invasive mechanical ventilation.

Conclusions: This case documents an unusual presentation of Ludwig's angina with multiple foci of infection and associated bone involvement.

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Osteomielitis mandibular como complicación de la angina de Ludwig: informe de un caso

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RESUMEN

Introducción: La angina de Ludwig es una celulitis gangrenosa de rápida progresión que afecta los espacios submandibular, sublingual y submentoniano. Su tratamiento estándar se basa en tres pilares: asegurar la vía aérea, terapia con antibióticos de amplio espectro y drenaje quirúrgico. La coexistencia de focos infecciosos activos simultáneos —un absceso submandibular profundo y osteomielitis mandibular— junto con la resolución del absceso sin drenaje quirúrgico constituye una presentación excepcional, raramente documentada en la literatura médica.

Caso clínico: Un paciente varón de 48 años con antecedentes de hipertensión presentó un cuadro clínico de una semana de evolución caracterizado por dolor en la mandíbula inferior izquierda, una masa submandibular izquierda, odinofagia y secreción purulenta en la base de la lengua, con deterioro progresivo que derivó en insuficiencia respiratoria aguda que requirió intubación orotraqueal. Se diagnosticó angina de Ludwig con un absceso que causó colapso de la vía aérea, junto con osteomielitis mandibular confirmada intraoperatoriamente durante una extracción dental y desbridamiento. Se indicó terapia antibiótica intravenosa, junto con un tratamiento conservador sin drenaje quirúrgico del absceso submandibular. Se observó una resolución progresiva del absceso submandibular durante el tratamiento, con extubación exitosa al sexto día de ventilación mecánica invasiva.

Conclusiones: Este caso documenta una presentación inusual de la angina de Ludwig con múltiples focos de infección y afectación ósea asociada.

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1. INTRODUCTION

Ludwig's angina is a rapidly progressing gangrenous cellulitis that simultaneously affects the submandibular, sublingual, and submental spaces of the floor of the mouth. It was first described in 1836 by the German physician Wilhelm Friedrich von Ludwig, and its term "angina" derives from the Latin angere, meaning "to strangle," in reference to its most serious complication: airway obstruction [1]. Ludwig's angina is a difficult-to-manage condition due to its rapid progression and the difficulty in maintaining a patent airway despite advances in antibiotic therapy, imaging techniques, and anesthetic management, with a mortality rate from airway obstruction ranging from 8–10% [2].

It is considered a rare but potentially fatal condition [3]. A retrospective analysis of emergency department admissions in the United States between 2006 and 2014 identified 5,855 patients diagnosed with Ludwig's angina. Of these, 47.2% required surgical drainage, and 4.6% required non-surgical management [4]. However, in more recent series of patients requiring admission to the intensive care unit (ICU), morbidity and mortality remain high at 50%, due to the difficulty in managing the airway [5]. Additionally, a multicenter study published in 2023 showed that, during the onset of the COVID-19 pandemic, the incidence of Ludwig's angina doubled compared to the pre-pandemic period (0.04 vs. 0.02 cases per year), and was also associated with a higher progression to necrotizing fasciitis and a longer duration of mechanical ventilation [6].

Ludwig's angina is characterized by contiguous spread of infection through the cervical fasciales planes. The infection can spread to the upper mediastinum, spreading directly along the fascial planes rather than through the lymphatic system. When Ludwig's angina progresses to necrotizing mediastinitis, negative intrathoracic pressure facilitates bacterial spread, significantly increasing morbidity and mortality [7].

Its primary cause is odontogenic; approximately 90% of cases originate from dental infections, primarily involving the lower second and third molars, whose roots penetrate the mylohyoid ridge, providing direct access to the submandibular space [8]. The most commonly reported predisposing factors are dental caries, recent dental procedures, diabetes mellitus, malnutrition, alcoholism, and

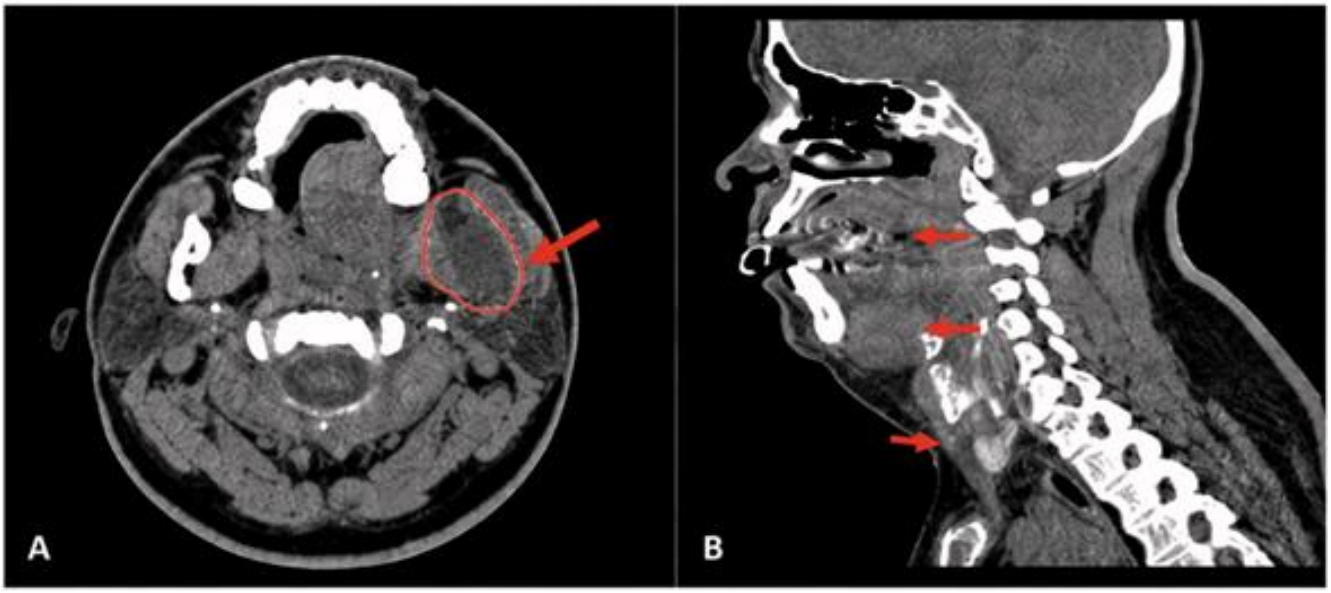


Figure 1: *Contrast-enhanced computed tomography of the neck. A: axial view showing an abscess in the left submandibular region; B: sagittal view showing diffuse soft-tissue density changes and contrast enhancement.*

immunocompromise, as in the case of patients with HIV infection and transplant recipients [9].

Recognizing the characteristic clinical presentation and early diagnosis are critical for prognosis. Stridor, dyspnea, dysphagia, sialorrhea, and blurring of the mandibular angle are common findings that should raise suspicion of Ludwig's angina [10]. Other signs include a mass in the submandibular space, a "double tongue" appearance, fever, and changes in the voice [11]. Diagnosis is confirmed by contrast-enhanced cervical computed tomography (CT), which allows for delineation of the extent of the infection and identifies collections that can be drained surgically. Airway management should not be delayed while awaiting imaging results when there is evident respiratory compromise.

Historically, management has been based on three fundamental pillars: securing the airway, broad-spectrum antibiotic therapy, and surgical drainage, which has been considered the most important intervention in treatment; however, this paradigm has begun to be questioned in the current literature. A series of 29 patients reported that 72% were managed conservatively following the initial course of the disease, with favorable outcomes in most of them [12]. Similarly, comparative studies have documented that in selected cases, resolution with antibiotics as the sole intervention is possible, although it is associated with a higher risk of airway compromise compared to the surgical approach [13]. In this context, spontaneous resolution of the abscess with medical management alone in a patient with severe Ludwig's angina who has already required orotracheal intubation, ICU admission, and the development of osteomyelitis constitutes a rare outcome of considerable academic value.

2. CASE REPORT

A 48-year-old male patient with a history of uncontrolled hypertension and an allergy to diclofenac presented with symptoms that had been present for one week, consisting of pain in the left lower jaw, which was initially managed on an outpatient basis with NSAIDs. Due to worsening symptoms and the appearance of a left submandibular mass with purulent discharge at the base of the tongue, he sought care at the clinic. A neck CT scan was performed, revealing acute tonsillitis associated with a 7-cc left submandibular abscess and reactive-appearing lymph nodes (Figure 1). Antibiotic therapy with intravenous ampicillin-sulbactam was prescribed. During his hospital stay, the patient developed type IV acute respiratory failure, requiring orotracheal intubation and invasive mechanical ventilation (IMV). A follow-up neck CT scan showed a submandibular abscess extending to the left paramedian inframandibular region, airway collapse, and displacement of adjacent structures. Consequently, the patient was transferred to the intensive care unit (ICU).

Antibiotic therapy was adjusted to piperacillin-tazobactam every 6 hours via continuous infusion, combined with clindamycin every 6 hours. The head and neck surgery team evaluated the patient, which decided to continue conservative management and serial ultrasound follow-up due to improvement in SIRS, reduction in facial and neck

edema, and laboratory improvement in acute-phase reactants such as CRP and white blood cell count. The patient gradually improved over time without requiring surgical drainage. On the sixth day of hospitalization, the patient was successfully extubated.

The dentistry and maxillofacial surgery departments evaluated the patient, which performed an intraoral examination revealing a gap in the lower left second molar with a poorly fitting metal inlay; the tooth was decayed with internal leakage, which was identified as the source of the infection. Consequently, extractions and bone debridement were performed. During the surgical procedure, bone involvement and a left submandibular fistula were observed, leading to a clinical diagnosis of mandibular osteomyelitis. Consequently, the decision was made to continue antibiotic treatment for 8 weeks at home, resulting in subsequent improvement and clinical resolution.

3. DISCUSSION

This case report describes a presentation of Ludwig's angina with characteristics that distinguish it from most cases reported in the literature. Resolution of the submandibular abscess through intravenous antibiotic therapy without the need for direct surgical drainage; the intraoperative finding of mandibular osteomyelitis as an associated complication identified during the extraction of the affected molars; and successful extubation after six days of invasive mechanical ventilation without the need for a tracheostomy. These elements constitute an uncommon clinical course of academic relevance in the context of deep neck infections.

The coexistence of Ludwig's angina, deep submandibular abscess, and mandibular osteomyelitis in the same patient and within the same clinical episode represents an uncommon progression. The literature describes osteomyelitis as a rare complication of severe odontogenic infections, with a reported incidence of 1 to 2 cases per 100,000 inhabitants per year in high-income countries [14]. In the specific context of Ludwig's angina, documentation of concomitant osteomyelitis as a third infectious focus was not systematically reported in the reviewed series, which gives this case particular reporting value. The intraoperatively documented fistula to the left mandibular suggests an extension of the infectious process that transcended the soft tissues and involved the bone and adjacent structures, likely facilitated by the delay in specialized care and the initial underestimation of the condition [3].

Surgical drainage has historically been considered the most important therapeutic pillar in Ludwig's angina. However, over the past decade, questions have arisen as to whether it is mandatory. Boscolo–Rizzo et al. reported that 72% of 29 patients were managed conservatively with satisfactory results [15]. The criteria that appear to guide the selection of conservative management include clinical stability, the absence of an organized collection, and an early response to antibiotic therapy, as in the present case, in which surgical drainage was not performed; instead, conservative management was maintained during the ICU hospital stay until spontaneous drainage occurred, thereby definitively resolving the infection. This finding reinforces the importance of active monitoring as a tool to guide therapeutic management [15].

Intraoperative documentation of osteomyelitis led to the extension of treatment to eight weeks of intravenous therapy, in accordance with recommendations for odontogenic osteomyelitis, which suggest treatment regimens lasting four to eight weeks, particularly when complete eradication cannot be guaranteed [9].

Contrast-enhanced computed tomography was decisive both in the initial diagnosis and in assessing the extent of the disease. However, serial ultrasound proved to be the follow-up tool that guided the decision not to perform surgery [10]. This finding of the absence of a collection supports the complementary use of both imaging modalities in the dynamic follow-up of Ludwig's angina. It should not be forgotten that in Ludwig's angina, which may coexist with mandibular osteomyelitis as a third active infectious focus, surgical drainage is not mandatory in all cases and that antibiotic therapy may be an effective strategy [7].

4. CONCLUSIONS

We present a case of Ludwig's angina with a documented triple site of infection that resolved without surgical drainage, using only broad-spectrum intravenous antibiotic therapy and serial imaging follow-up. This case challenges the paradigm of surgical drainage as a mandatory intervention and documents mandibular osteomyelitis as a rarely described concomitant complication in this context. Case series are needed to more precisely define the selection criteria for conservative management in Ludwig's angina with multicompartmental involvement.

5. CONFLICT OF INTERESTS

The authors have no conflict of interest to declare. The authors declared that this study has received no financial support.

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